



Today's agenda

- ↳ How to approach a problem.
- ↳ Design a Pen.



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↳ Types of LLD/loops interviews

1. Theoretical: OOPs knowledge

2. Design a system like: booking show, Splitwise, elevator etc.
HLD based with some LLD concepts

↳ class diagram → {45-60 mins}

↓
No Coding

↳ Big tech giants {Amazon, ms etc.}

3. Machine Coding

↳ designing + working code ~ 2-3 hours

↳ Big startups: Phonepe/walmart/zomato/Paytm etc.



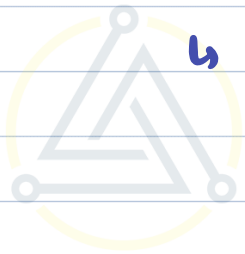
↳ Machine coding { 2-5 hours }

↳ clarify requirements
↳ use case diagram
↳ class diagram
↳ Schema designxx

} 45-50 mins

↳ implement the code] 1 hr 30 mins

↳ Discussion on code] 10-15 mins



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Design a Pen

↳ Anything that can write is a pen.

1. Different types of Pen { ball, Gel, fountain }
2. Pen might have or haven't a refill.
3. Refill will have ink.

↳ ink can be of different color or type.



4. Refill has a nib. For fountain pen, pen will directly have a nib.

5. Every pen will have brand, cost & weight.

6. Some kind of pen can be refilled but some can't be.

→ write() → every pen has different way of writing.
→ smooth way
→ medium way
→ never use way

→ refilling() → How can we refill the ink.

↳ different way of writing: Strategy design pattern.

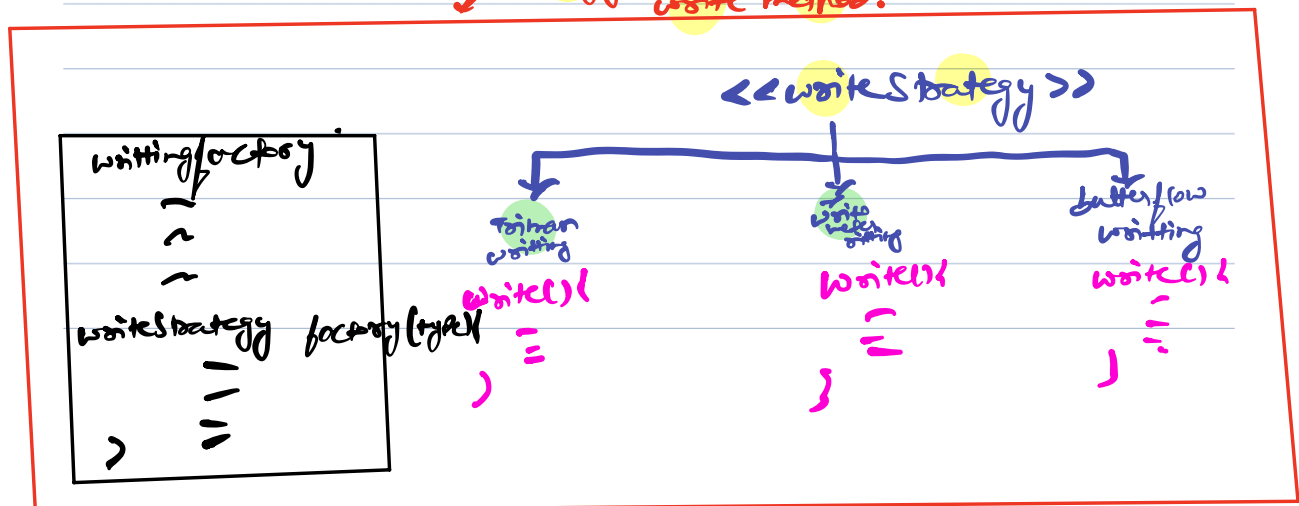
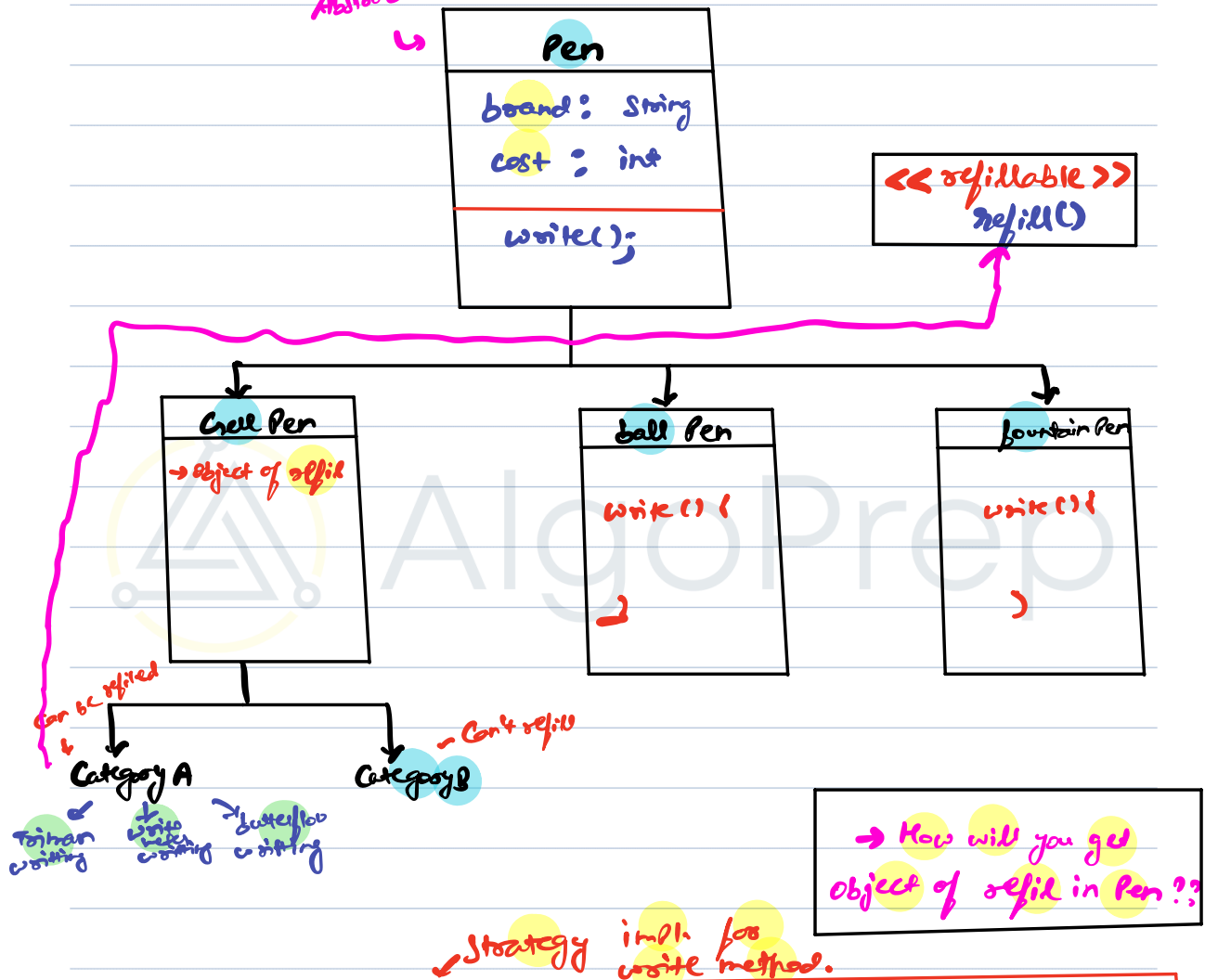
↳ based on the input, we will create a pen: Factory design pattern.



Class diagram

↳ Nouns in requirements.

↳ Abstract class





Refill
Ink Nib

Ink
- color - list of features

nib
~ radius ~ length



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